

Assignment 4 — Hyperparameter Tuning Report

English $\rightarrow$ Hindi Neural Machine Translation (Transformer)

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1 Baseline Metrics (100 Epochs)

Metric	Value
Total Epochs	100
Final Training Loss	0.0917
BLEU Score (Validation)	52.47
Total Training Time	85.17 minutes

Baseline Model: Transformer with $d_{model} = 512$, 6 layers, 8 heads, $d_{ff} = 1024$, dropout=0.1, learning rate= 10^{-4} , batch size=64, Adam optimizer.

2 Hyperparameter Search Space

Ray Tune with Optuna search (OptunaSearch) and ASHA early-stopping scheduler was used. 15 trials were run; each trial trained for up to 25 epochs.

Hyperparameter	Search Range / Choices
Learning Rate (lr)	Log-Uniform $10^{-5} \rightarrow 10^{-1}$
Batch Size	{16, 32, 64}
Attention Heads (num_heads)	{4, 8, 16, 32}
Feed-Forward Dim (d_{ff})	{1024, 2048}
Dropout	Uniform 0.1 $\rightarrow$ 0.5
Optimizer	{adam, adamw}
Weight Decay	Log-Uniform $10^{-5} \rightarrow 10^{-2}$

3 Best Configuration Found

Hyperparameter	Best Value
Learning Rate	0.000147
Batch Size	32
Attention Heads	32
Feed-Forward Dim (d_{ff})	2048
Dropout	0.154
Optimizer	Adam
Weight Decay	1.96×10^{-5}
Epochs (sweep trial)	25

4 Final Tuned Model Metrics

Metric	Baseline (100 ep)	Tuned (25 ep)	Improvement
Training Loss	0.0917	0.1538	0.014
BLEU Score	52.47	69.39	+16.92 pts
Training Time	85.17 min	~15 min	~5.7× faster
Epochs to Beat Baseline	100	25	75% fewer